

Study number: 20111212	Programme: MSc2
Evaluation subject: Optimisation Theory and Modern Reliability from a Practical Approach Friday 9 June 2017 at 9:00 – 13:00	

Please write your study no. on all pages. Do not write your name as your evaluation is anonymous!

Total number of pages, including this page: 20

Please, only write on one side of the papers that you hand in.

NB! Your paper must be easy to read. If this is not the case, your paper may be evaluated as “not passed”.

*All usual aids are allowed (notes, books, tables, calculator and PC).
You are not allowed to communicate amongst each other or with the outside world which means that the use of mobile phone, Wi-Fi, internet, email is not allowed.*

You are allowed to take the examination questions with you. But you are NOT allowed to take them with you if you leave the room before the examination has ended.

EX: 06 (a)

The weibull plots for the failure of four (4) individual group as shown fig. 3.

The voltage level

Time	voltage
1000	4.0
10.5	5.0
1.8	5.5
0.01	6.0

Since the β weibull Plot for the case of 6V voltage is significantly different from others, and time is also less (0.01). The failure mechanism under 6V testing condition is considered as different, which

(2)

Can not be used for the lifetime model to be derived from the other three groups of life data.

The figure 3 shows three plots (B) with similar B values approximately -

EX: 06 (b)

Empirical Lifetime Models:

→ Arrhenius model $AF = \exp \left[\frac{E_A}{K} \left(\frac{1}{T_{field}} - \frac{1}{T_{test}} \right) \right]$

$E_A \rightarrow$ activity energy for the process.

$K \rightarrow 8.62 \times 10^{-5} \text{ eV/K}$ (Boltzmann constant)

T_{field} and $T_{test} \rightarrow$ absolute Kelvin temperature at field and test level respectively.

$$AF = \left(\frac{RH_{test}}{RH_{field}} \right)^m \exp \left[\frac{E_A}{K} \left(\frac{1}{T_{field}} - \frac{1}{T_{test}} \right) \right]$$

$m \rightarrow$ humidity power system
(range 2.0 and 4.0)

$RH \rightarrow$ Relative humidity measured as %.

CON $\rightarrow$ EX 6 (b)

$$AF = \left(\frac{V_{\text{Test}}}{V_{\text{field}}} \right)^B$$

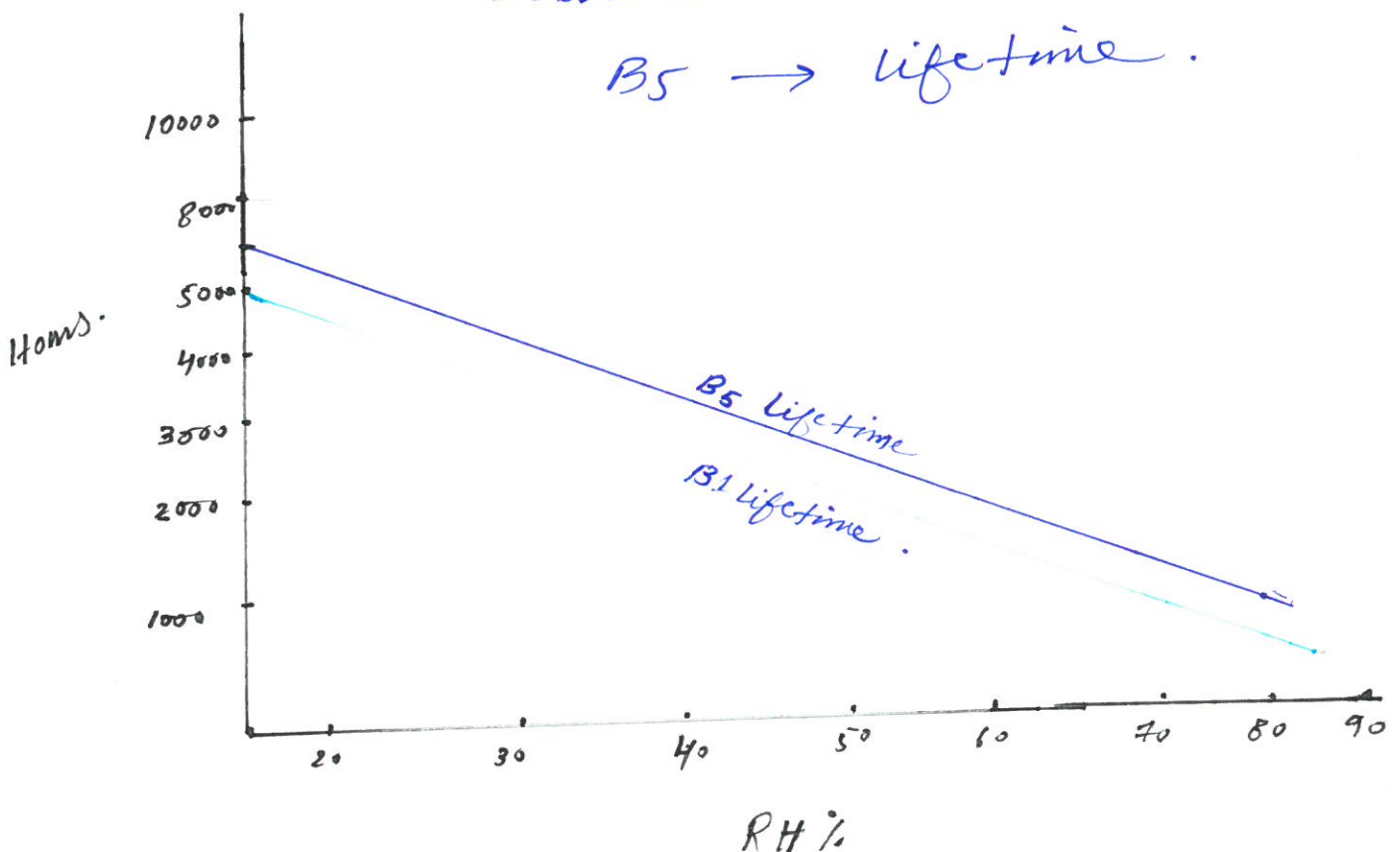
An alternative exponential voltage model takes the form of:

$$AF = \exp [B(V_{\text{Test}} - V_{\text{field}})]$$

Bx $\rightarrow$ life time versus RH% $\rightarrow$ approximate Plot.

Weibull $\rightarrow$ 50% confidence.

B5 $\rightarrow$ lifetime.



Ex: 5 CONT:—

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Q5

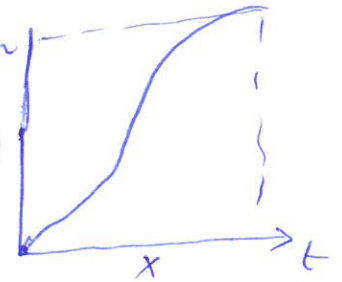
→ Weibull Distribution.

Swedish Scientist

$$f(t) = \left\{ \frac{\beta}{\eta^\beta} t^{\beta-1} \exp \left[- \left(\frac{t}{\eta} \right)^\beta \right] \right\}^{t \geq 0}$$

There are two parameters

$\eta \rightarrow$ either $\beta \rightarrow$ Better for



Results Analysis.

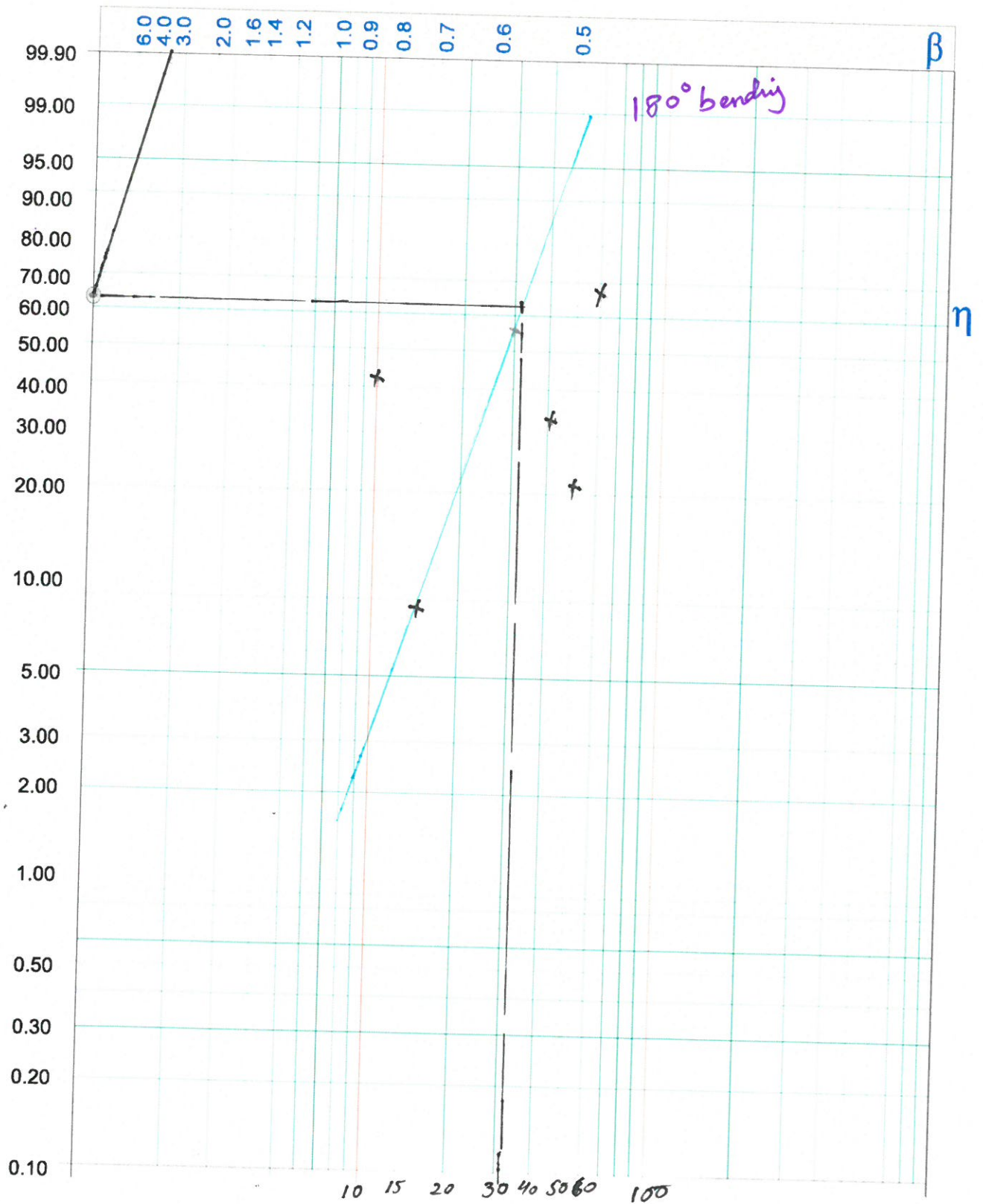
c) is (d).

(b)

180°	
η	30
β	3.6
cycles	30% fail

	Median rank %	Cycles (180°)	
1	8.3%	15	
2	20.21%	50	
3	32.13%	40	
4	44.04%	10	
5	55.96%	30	
6	67.87%	60	
7	79.79%	—	
8	91.7%	—	

$$\beta_{180} = 3.6$$



$$\eta_{180} = 30$$

EX: 5

④

a)
$$F(t) = 1 - \exp \left[- \left(\frac{t}{\eta} \right)^{\beta} \right]$$

b) See table (attached).

c) $\beta = 3.6$ $\eta = 30$. These ranges can be acceptable due to the error of the curve fitting.

d) β is obtained by using Parallel translation of the fit line to the β -normogram in the Weibull plot. As shown in Weibull expression, regardless of the variation of β , $F(t)$ becomes 63.2% if t equals η . That is why η refers to the cycles when failure reaches ~~62.3%~~ 62.3%.

EX: 8

⑤

a)

Target Data T.

Temperature range stress level	Nr. cycles	Point
$90-70=20^\circ$	5	Block A
$90-40+20-30=20$	15	Block B
$90-50+30-10=30^\circ$	5	Block C
$90-90=0$	0	Block D

b)

$$N_f = A \cdot \Delta T^\alpha$$

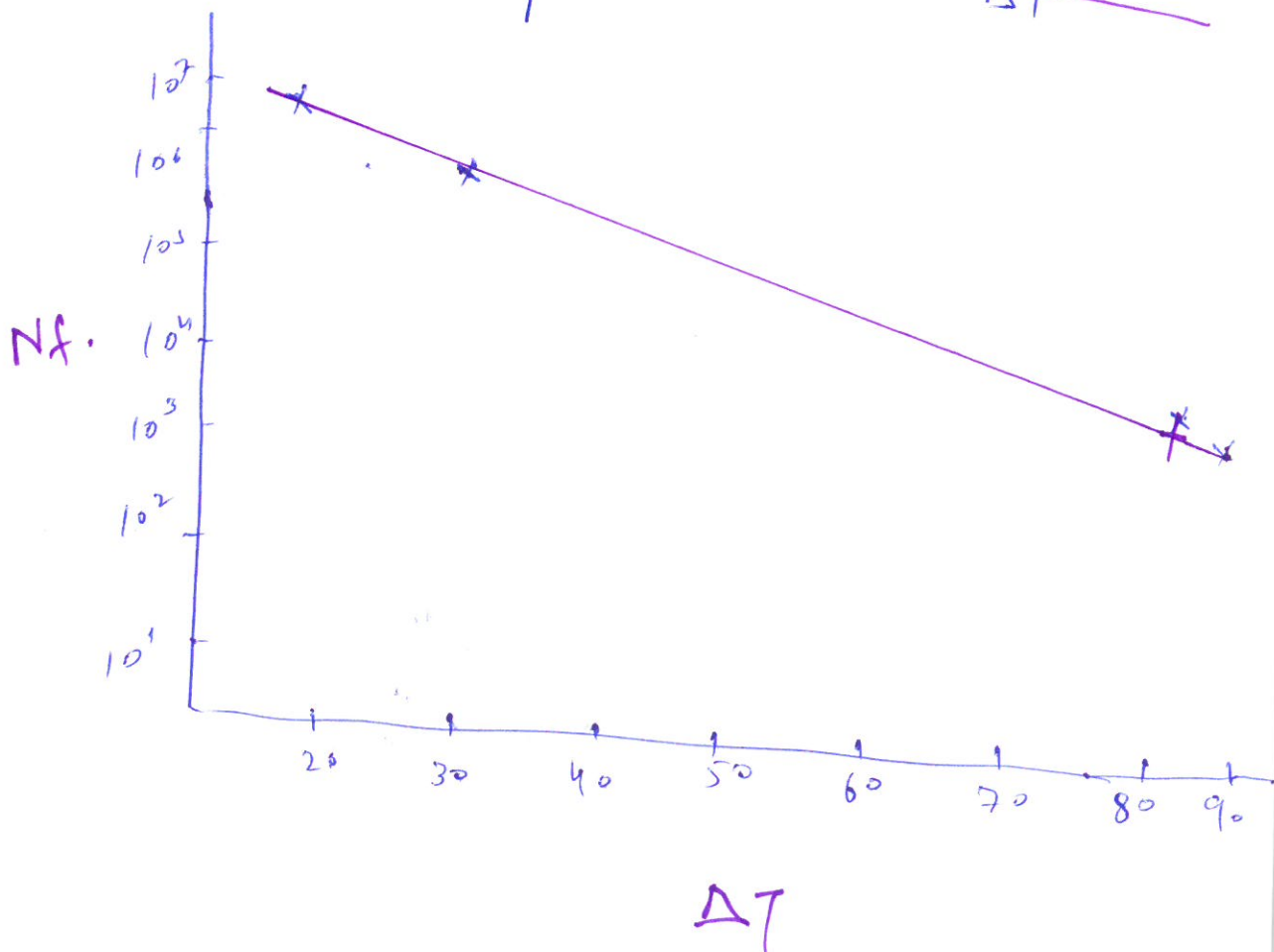
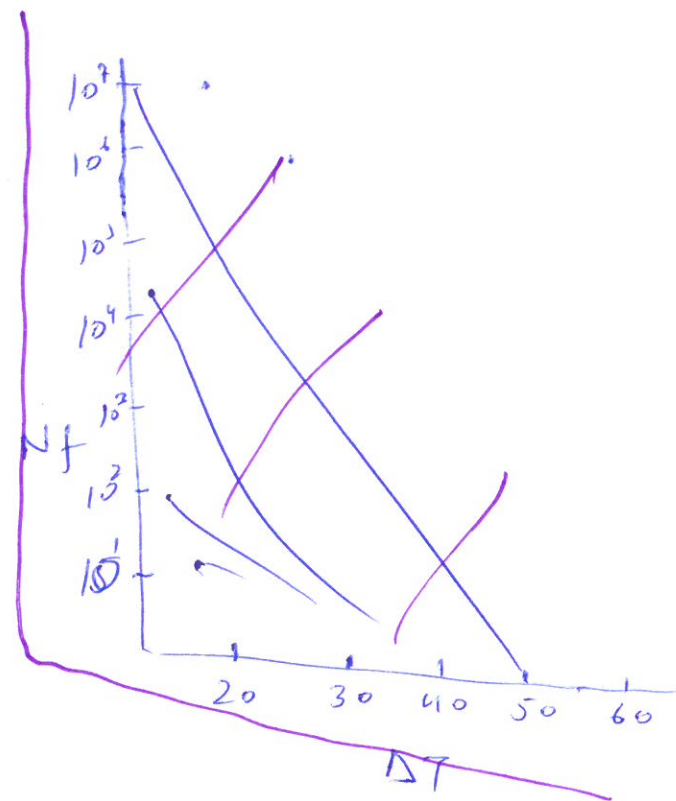
$$A = 1.99 \times 10^{14}, \alpha = -5.039$$

(P.T.O)

b) Contin:

⑥

ΔT	N_f
20	5.5320×10^7
30	7.1720×10^6
40	1.6830×10^6
50	5.4659×10^5
60	2.1815×10^5
70	$1.0032 \cdot 10^5$
80	$5.1190 \cdot 10^4$
90	$2.8276 \cdot 10^4$



b) cont:

(7)

$$A_{DN} = \frac{N_1}{NF_1} + \frac{N_2}{NF_2} + \dots + \frac{N_n}{NF_n}$$

$$= \frac{5}{5.533 \times 10^7} + \frac{15}{\cancel{5.533 \times 10^6} \times 5.533 \times 10^7} + \frac{5}{7.173 \times 10^6} + 0$$

$$= 9.036 \times 10^{-8} + 2.71 \times 10^{-7} + 6.97 \times 10^{-7}$$

$$A_{DN} = 1.058 \times 10^{-6} \text{ hours.}$$

c)

Expected life time in one year.

$$\text{Lifetime (Year)} = \frac{T_{\text{period}}}{A_{DN}} = \frac{8760}{1.058 \times 10^{-6}}$$

$$\text{Life time} = 8279 \times 10^{-6}$$

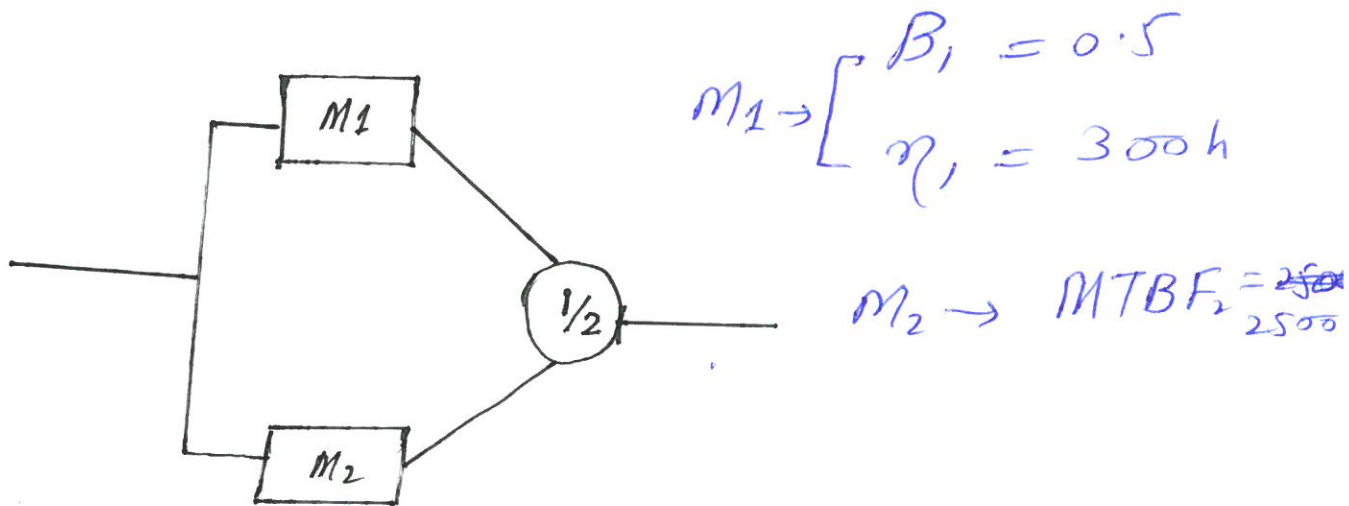
Ans.

EX:- 7

⑧

a)

RBD Schematic



- Independent components
- Each has reliability R .

The system works properly if at least m of n works properly. Therefore, the reliability is

$$R_{sys} = 1 - \sum_{i=0}^{m-1} \binom{n}{i} R^i (1-R)^{n-i}$$

exponential case:

$$R_{sys} = 1 - \frac{1}{(\lambda t + 1)^n} \sum_{i=0}^{m-1} (\lambda t)^{n-i}$$

b) EX-7 cont:—

(9)

$$\beta_1 = 0.5 \quad \eta_2 = 3000 \text{ h.}$$

$$\eta_2 \rightarrow MTBF_2 = 2500$$

$$T_{\text{trim}} = 200 \text{ h.}$$

Exponential

$$R(200) = \exp\left(-\frac{200}{3000}\right)$$

$$= 0.9355$$

EX-01

90

Company Product Type A & B.

n_A = number of stores of A

n_B = number of stores of B.

Standard optimization model

$$\min f(n_A, n_B) = \frac{1}{500 - 900 + (150 - 500)n_A + (600 - 400)n_B}$$

alt under
brackets - ok

$$n_A \geq \frac{500}{20} \quad , \quad 5000 - n_A \leq 0$$

$$n_B \geq \frac{500}{24} \quad , \quad 5000 - n_B \leq 0$$

$$20n_A + 24n_B \leq 5000 \Rightarrow \checkmark$$

$$20 - n_A + 24n_B - 5000 \leq 0$$

EX-2

(14)

$$q) f(x) = (x_1 - 1)^2 + 2x_2^2$$

$$x^* = [3, 1]^T$$

$$f(x^*) = (3-1)^2 + 2 \cdot$$

$$f(x^*) = 6$$

The gradient

$$c^0(x) = \begin{bmatrix} 2x_1 \overset{-2}{(\cancel{x_1 - 1})} \\ 4x_2 \end{bmatrix} \quad \% 1$$

$$c^0 = \begin{bmatrix} 2 \\ 4 \end{bmatrix} \quad \% \quad 2 \cdot 3 = 6 \text{ or } 2 \cdot 4 = 8$$

Norm

$$\begin{aligned} \|c^0\| &= \sqrt{(2)^2 + (4)^2} \\ &= \sqrt{20} > \epsilon \end{aligned}$$

St. Nr. 20/112

Steepest descent

(12)

$$d^0 = -(\nabla f(x^0)) = -\begin{bmatrix} 4 \\ 2 \end{bmatrix} \quad \text{only}$$

$$x_1 \rightarrow (x_1 + \alpha d_1)$$

$$x_2 \rightarrow (x_2 + \alpha d_2)$$

$$\begin{aligned} f(\alpha) &= f(x^0 + \alpha d^0) = (x_1 + \alpha d_1)^2 - 1 + 2 \cdot (x_2 + \alpha d_2)^2 \\ &= (3 + \alpha \cdot 4)^2 - 1 + 2 \cdot (1 + \alpha \cdot 2)^2 \end{aligned}$$

$$f(\alpha) = 40\alpha^2 - 20\alpha - 3 \quad \text{Using Matlab for Calculat}$$

$$f'(\alpha) = 80\alpha - 20 \quad f'(\alpha) = 0$$

$$\boxed{\alpha = 1/4}$$

$$\delta = 0.05 \leftarrow ?$$

(12)

$$X^{(1)} = X^* + \alpha d^* = \begin{bmatrix} 3 \\ 1 \end{bmatrix} + 0.25 \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 0.5 \end{bmatrix} = \begin{bmatrix} 4 \\ 1.5 \end{bmatrix}$$

$$f(X') = (4-1)^2 + 2(1.5)^2$$

$$f_{\text{opt}} = 13.5$$

$$(f(X_1) = 13.5) \text{ Ans.}$$

3

(14) ~~10~~

$$b) (x_1 - 1)^2 + 2x_2^2$$

$$x^0 = [3 \ 1]^T$$

$$c^0 = \begin{bmatrix} 2x_1 \\ 4x_2 \end{bmatrix} \quad \checkmark$$

$$c^0 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

Norm

$$\|c^0\| = \sqrt{(2)^2 + (4)^2}$$

$$= \sqrt{20} > \in$$

$$H = \begin{bmatrix} 2 & 2 \\ 4 & 0 \end{bmatrix} \quad \checkmark$$

$$H^{-1} = \begin{bmatrix} 0 & 0.25 \\ 0.5 & 0.5 \end{bmatrix} \quad \text{using } \text{inv}(H)$$

Matlab

 $y = \text{inv}(H)$

$$d^0 = -H^{-1}c^0 = - \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

STNR. 2011/2/2
 (15)

$$d^0 = -H^{-1}c^0 = - \begin{bmatrix} 0 & 0.25 \\ 0.5 & -0.5 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

→ Modified Newton's method is always possible to calculate a search direction at any point if Hessian is % singular

→ Newton direction is always that of decent

2

EX-4.

15

a) Perotopical point (49, 49)

b) sheet attached.

c) see Please sheet attached.

d) (10, 10) utopia point. %

e)

$$U(X^*) = 2 \left[(X_1 - 5)^2 + 1 (X_2 - 25)^2 \right] + 1 \left[(X_1 - 15)^2 + (X_2 - 5)^2 \right]$$

$$w_1 = 2$$

$$s \ w_2 = 1$$

The comparison is $U = w_1 f_1(X) + w_2 f_2(X)$ % Constraint
 solution with lowest "U" is

located at a point on the Pareto front
 that is located the shortest distance
 from utopia point graph shows. %

utopia point (10, 10)

and perotopoint (49, 49) %

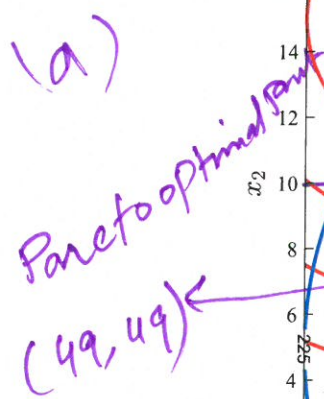
3

ST. NR. 20111212

~~16~~

17

SOLUTION


$$n_1 \leq 10$$


50
40
30
20
10
0

Ex-4
Cont

S. Nr. 2011/212

~~17~~
18

e) $N = 124$

So that U is between
14 ~~17~~ and 10 ~~now~~ it 12.

$$X^* = \begin{bmatrix} 10 \\ 12 \end{bmatrix}$$

$$U^* = 2 \left[(10 - 5)^2 + (12 - 15)^2 \right] + 1 \cdot \left[(10 - 15)^2 + (12 - 5)^2 \right]$$

$$U^* = 2 \cdot [25 + 9] + (-5 - 7)^2$$

$$U^* = 80 \text{ Ans.}$$

f) It is global optimum.

Q

EX-3

3.N/2011/2/2

(19)

$$\min f(x) = -5x_1 - 2x_2$$

Into dueng.

$$-x_1 + x_2 + x_3 = 10 \quad \checkmark$$

$$2x_1 - x_2 + x_4 = 20$$

$$x_1 \geq 0 \quad \forall i = \{1, 2, 3, 4\}$$

	x_1	x_2	x_3	x_4	b
x_3	1	-1	1	0	10
x_4	2	-1	0	1	20
	-5	-2	0	0	

1. Divide row 1 by 2. [?]

20/11/21 (20)

	x_1	x_2	x_3	x_4	
x_3	$-\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	0	$\frac{10}{2}$
x_4	2	-1	0	1	20
	-5	-2	0	0	

2. Subtract old row 1 from row 2

No time

✓

(1)